

Problem 3

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Problem 1 The limit of Riemann sums for a function f on the interval $[1, 5]$ is given by

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x \text{ on } [1, 5].$$

(a) Identify f and express the limit as a definite integral.

(b) Evaluate the limit of Riemann sums.

[4]